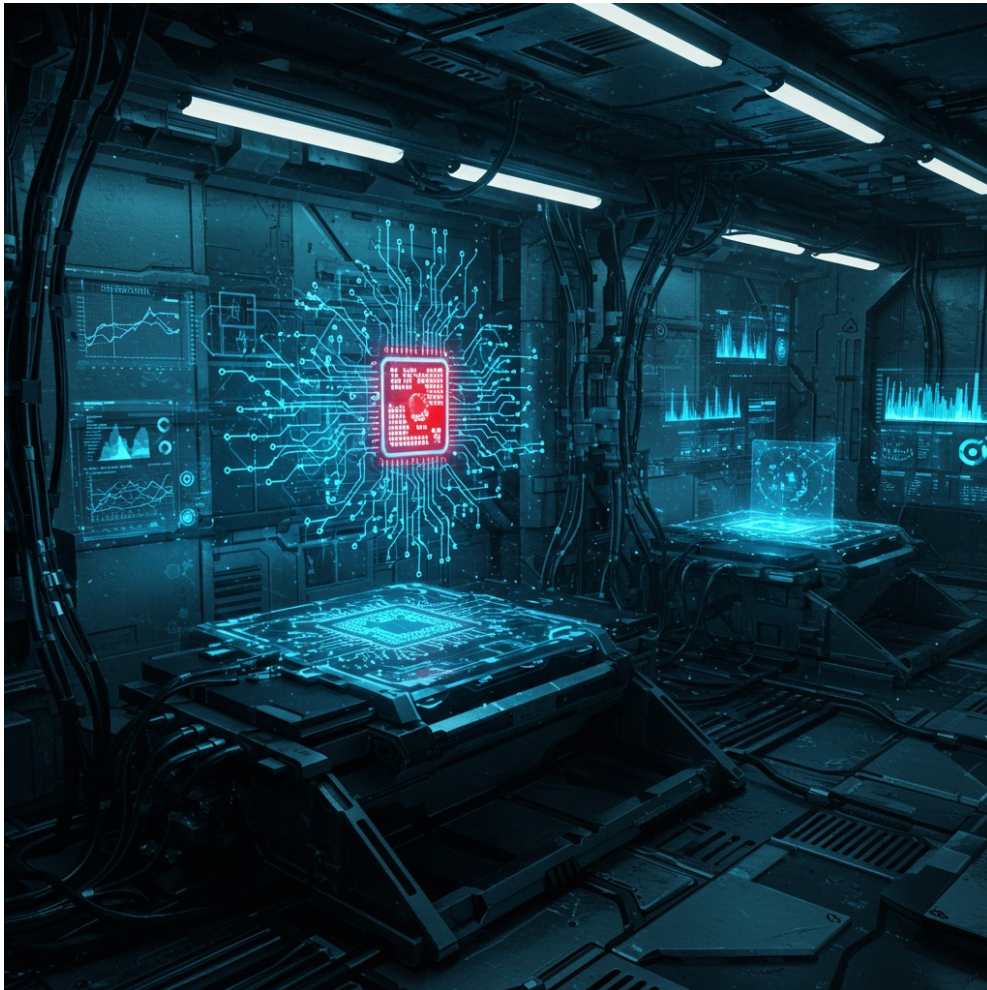
DETECCIÓN ANOMALIAS CON DEEP LEARNING



Valero Laparra

INTRO

- Predicción

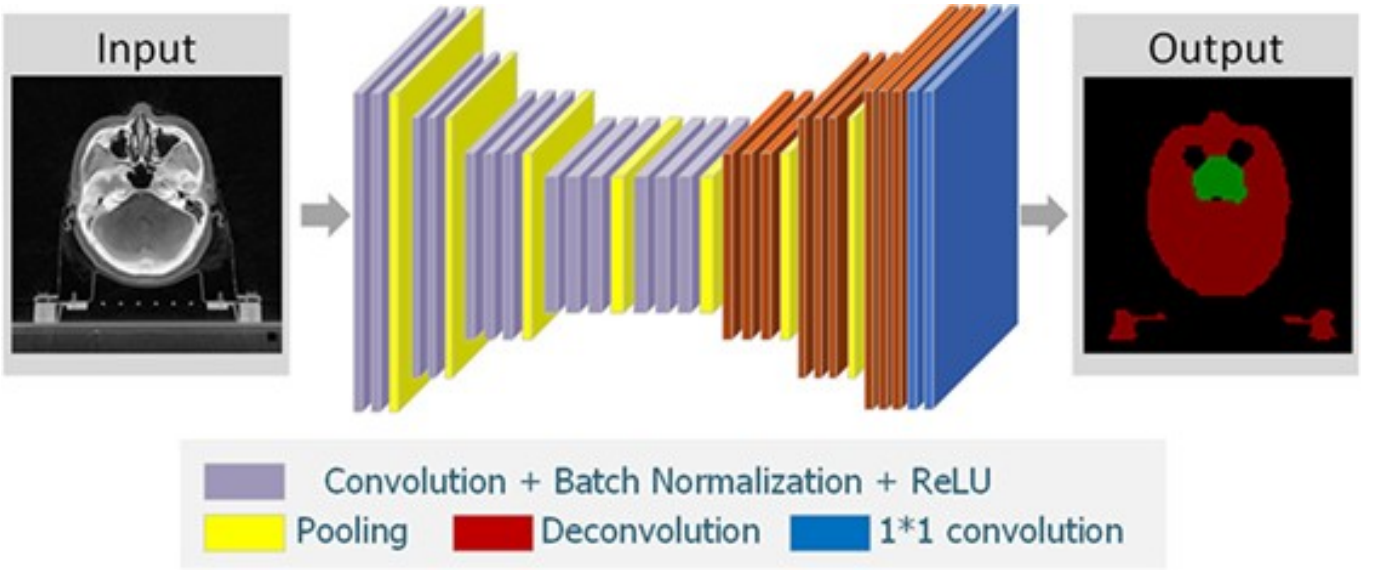
- Autoencoders

- Modelos generativos
    - Explícitos
    - Implícitos

Modelos predictivos

Supervisado

$$A = f(x)$$



Autoencoders

Definición:

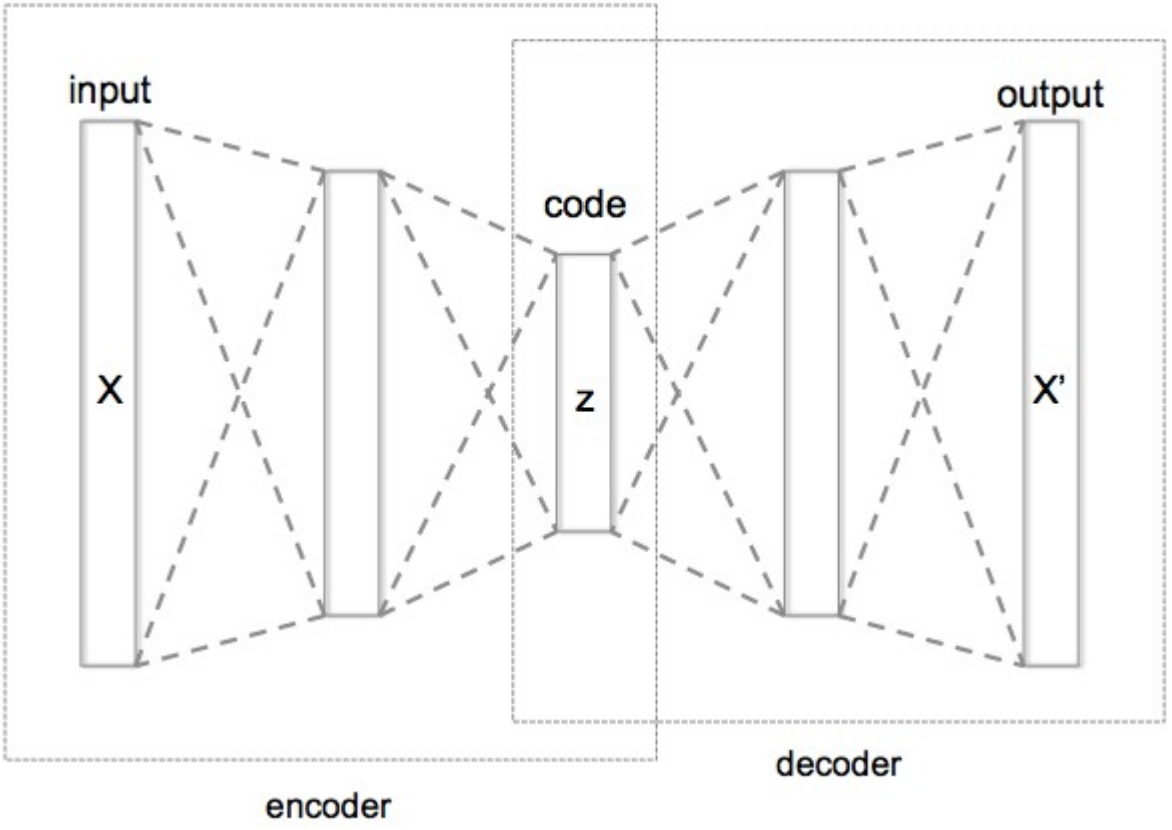
Z = enc(**X**)

X = dec(**Z**)

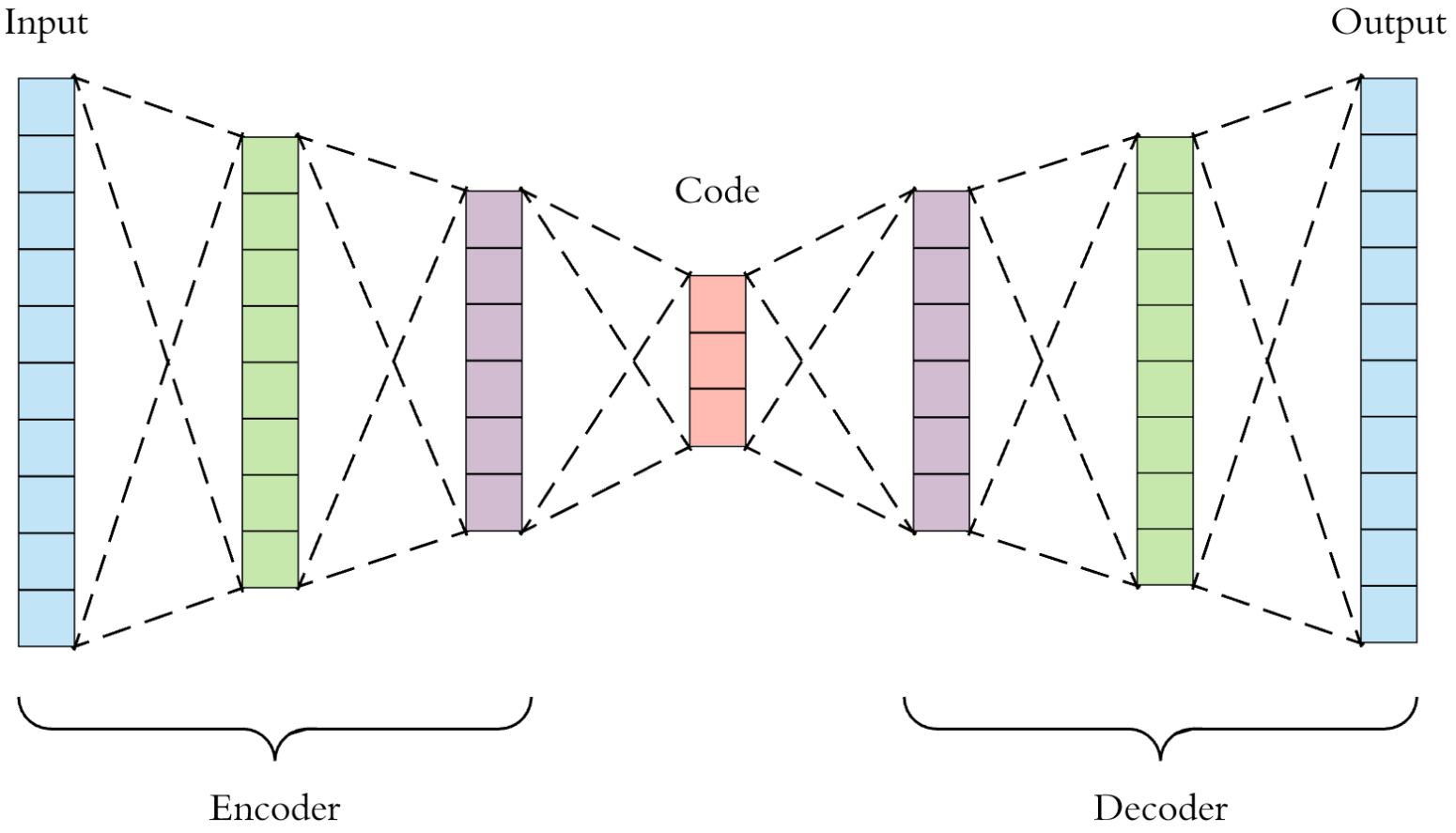
restricciones**(Z)**

No supervisados!

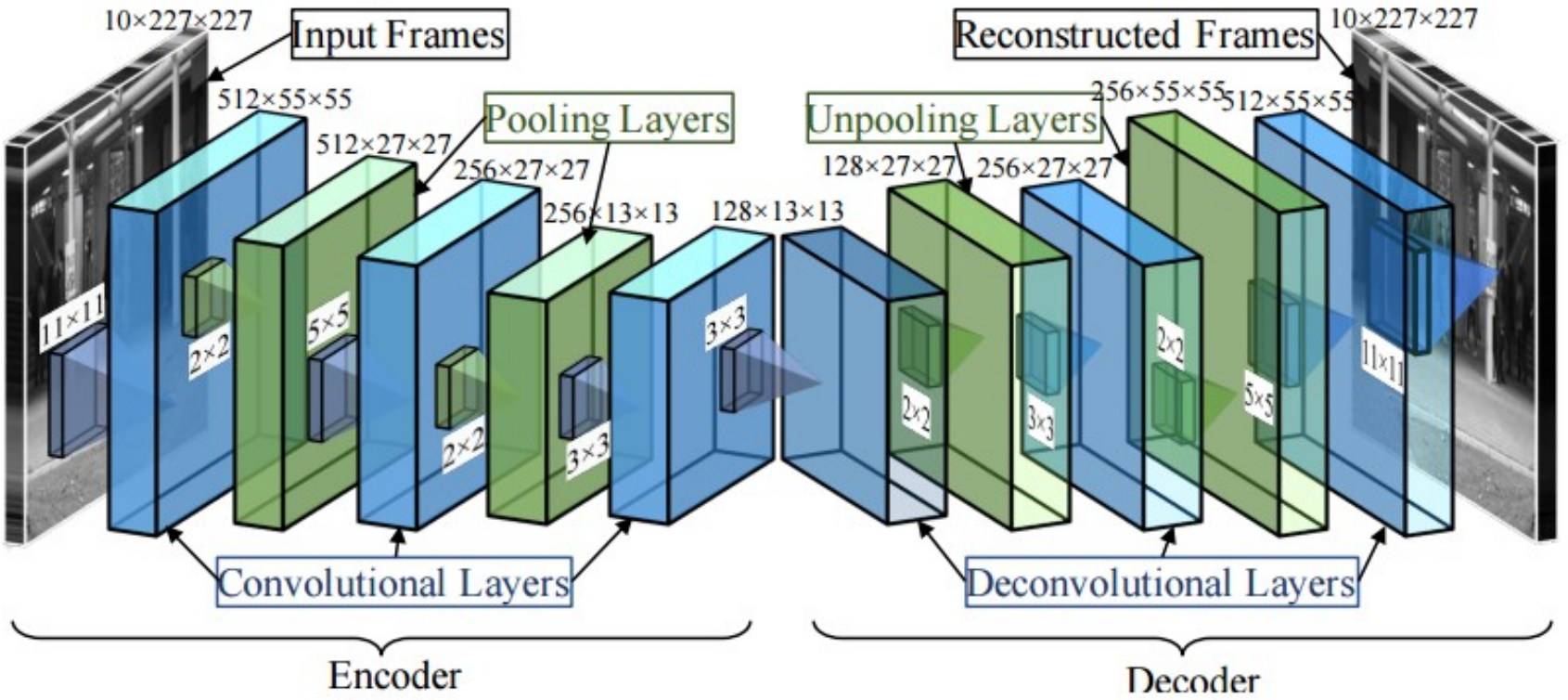
A(x) = |f(x)-x|



Deep AE



CNN AE
Convolutional

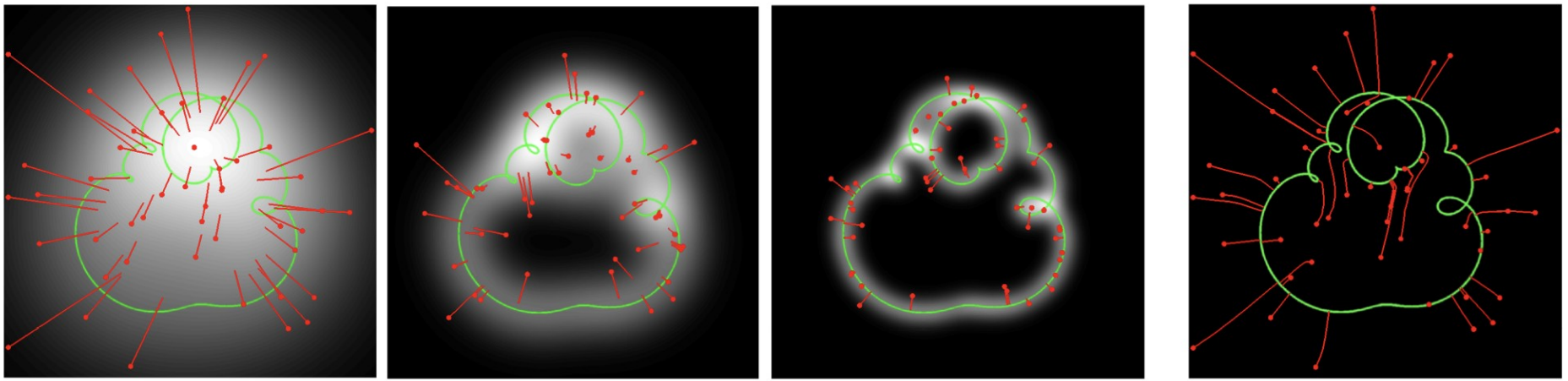


CODE: https://www.tensorflow.org/tutorials/generative/autoencoder

Denoising based

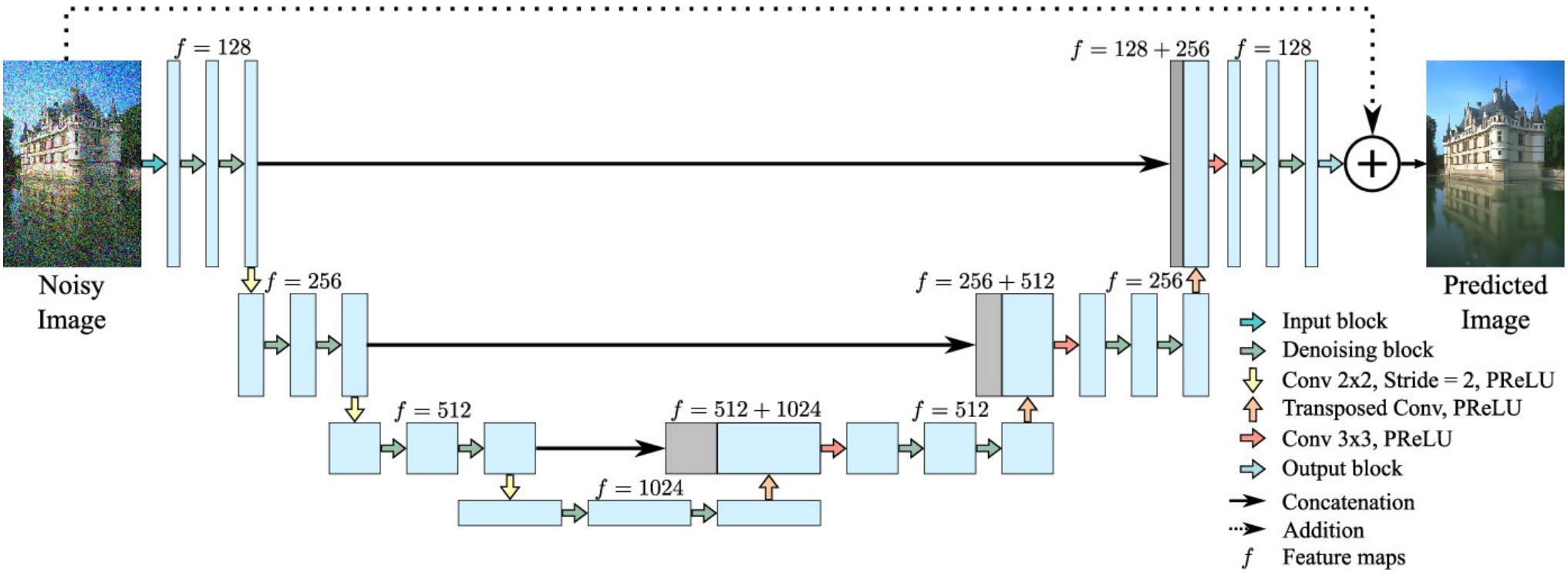
$$\hat{x}(y) = y + \sigma^2 \nabla_y \log p(y)$$

x-y ≈ Dp(x)



CNN AE – UNET

Para denoising
(normalmente para segmentación)



Modelos Generativos

- Explícitos:

    - Flows
    - Autorregresivos (predictivos)
    - VAEs

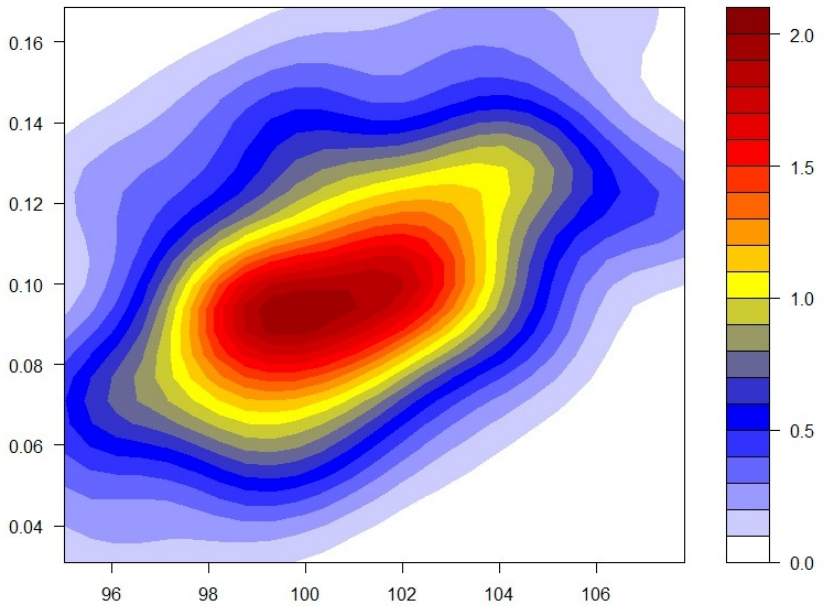
- Implícitos:

    - GANs
    - Diffusion

Modelos Generativos

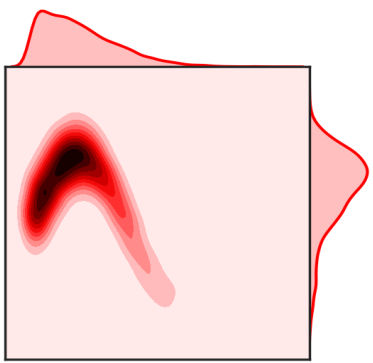
- **Explícitos**

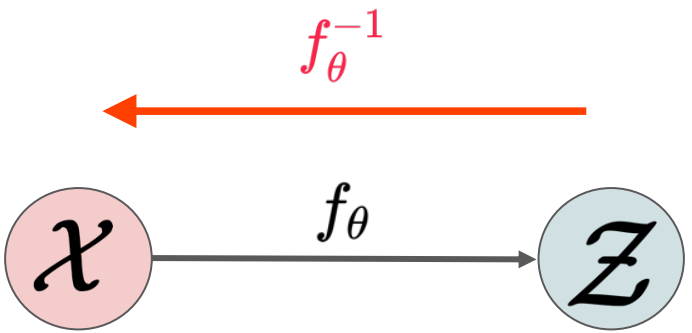
$$p(x) = f(x)$$



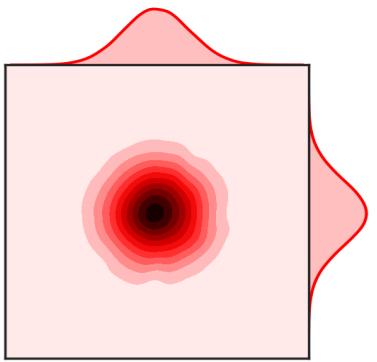
Normalizing Flows

Data Distribution



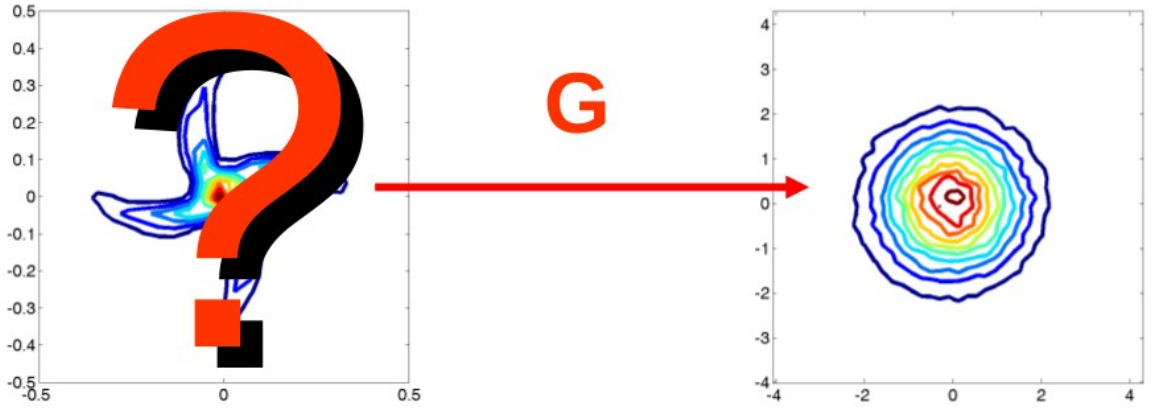


Known Distribution



Flow: Rotation Based Iterative Gaussianization (RBIG)

From P(x) to P(y) (Gaussian)

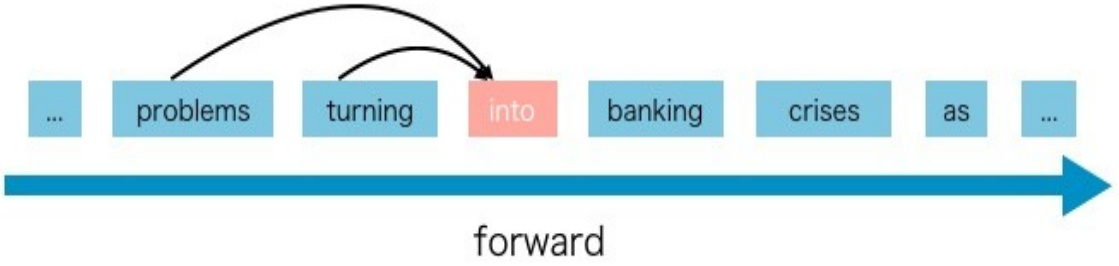


$$p_{\mathbf{x}}(\mathbf{x}) = p_{\mathbf{y}}(\mathcal{G}(\mathbf{x}))\left|\frac{d\mathcal{G}(\mathbf{x})}{d\mathbf{x}}\right|^{-1}$$

Autoregresive Generative Models

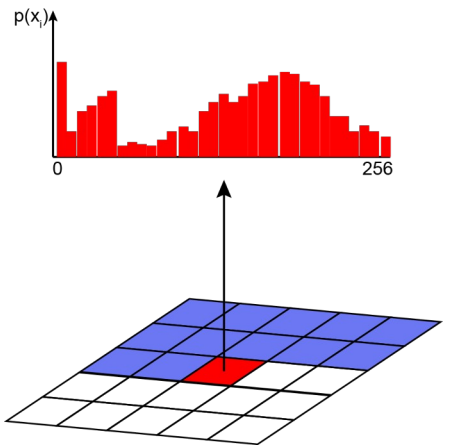
Texto

(LLMs)



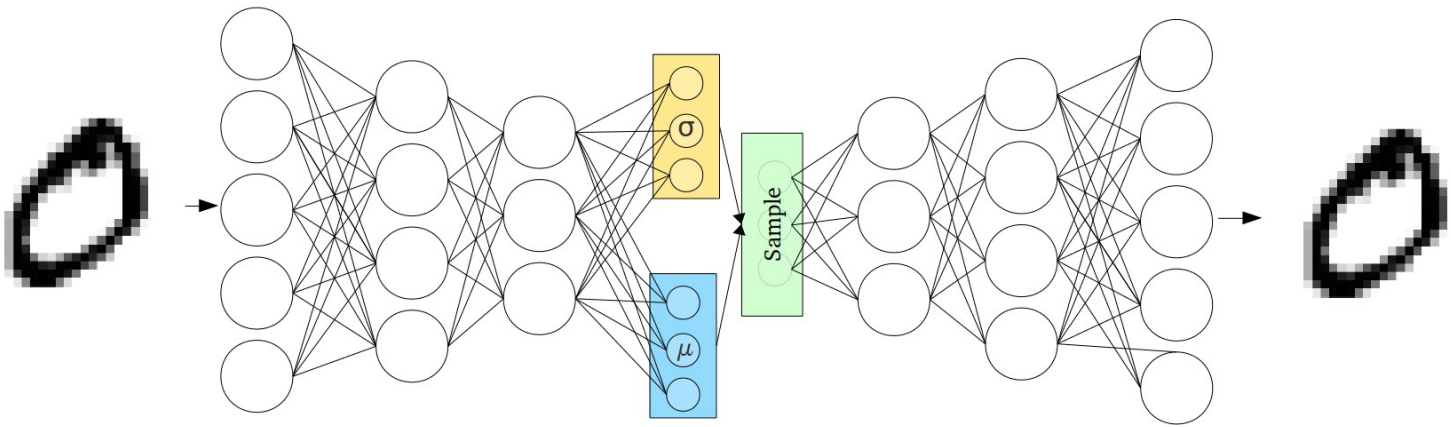
Imagen

(PixelCNN)



Variational AE

- El espacio latente es continuo
- Modelo generativo



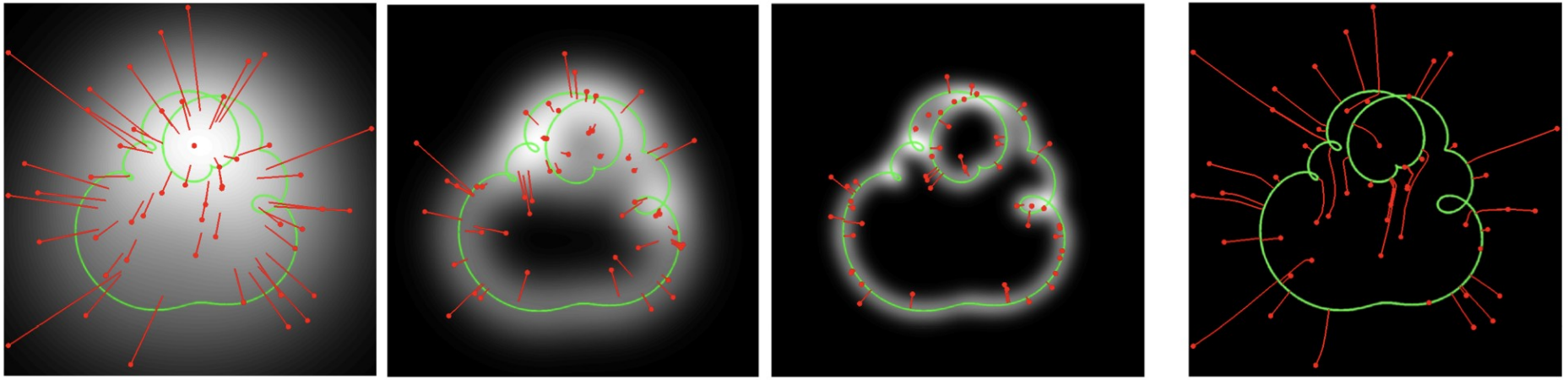
CODE: https://www.tensorflow.org/tutorials/generative/cvae

JUGAR: http://leonchen.io/keras-js/#/mnist-vae

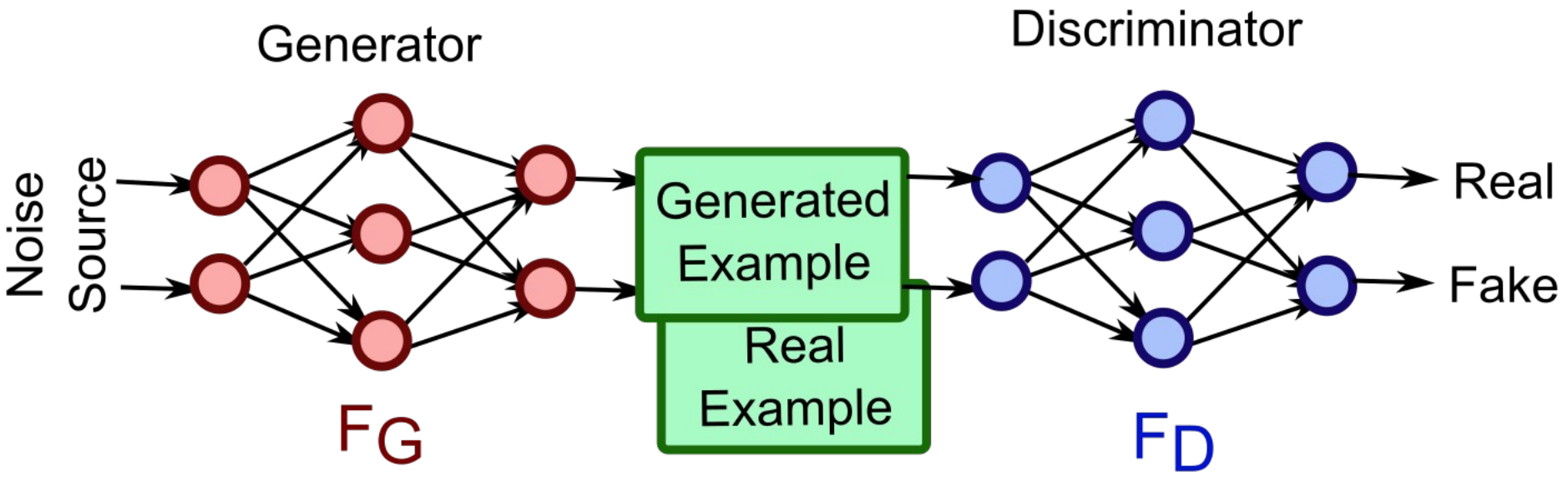
Modelos Generativos

- Implícito

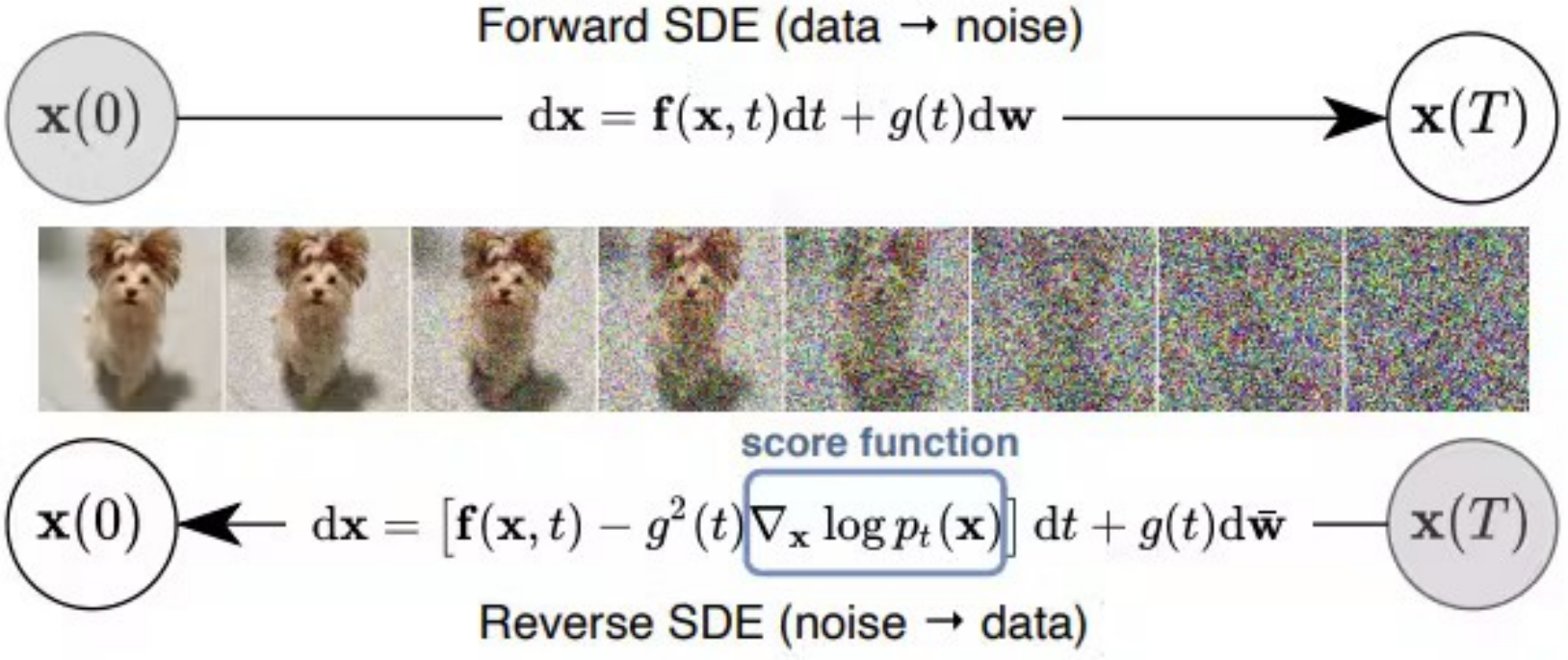
$$p(x) = g(f(x))$$



GANs


Generator
Discriminator
Noise Source
Generated Example
Real Example
Real
Fake
F_G
F_D

Diffusion


Forward SDE (data → noise)
$\mathbf{x}(0)$
$\mathrm{d}\mathbf{x} = \mathbf{f}(\mathbf{x}, t)\mathrm{d}t + g(t)\mathrm{d}\mathbf{w}$
$\mathbf{x}(T)$
score function
$\mathbf{x}(0)$
$\mathrm{d}\mathbf{x} = \left[\mathbf{f}(\mathbf{x}, t) - g^2(t)\nabla_{\mathbf{x}} \log p_t(\mathbf{x})\right] \mathrm{d}t + g(t)\mathrm{d}\bar{\mathbf{w}}$
$\mathbf{x}(T)$
Reverse SDE (noise → data)